

RITIK KUMAR

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PROFILE

Physics researcher at the Indian Institute of Science (IISc Bengaluru) working at the intersection of machine learning, statistical inference, scientific computing, and large-scale data analysis. Experienced in developing probabilistic and deep learning models for noisy time-series data, uncertainty quantification, Bayesian inference, and large-scale research pipelines. Strong mathematical foundations (JEE Advanced AIR OBC 3424, top 0.3%) combined with end-to-end ML pipeline ownership from data ingestion to statistical validation. Published author with international research collaborations across Japan, Germany, and India. Interested in quantitative research, machine learning, and data-driven decision making.

EDUCATION

Indian Institute of Science (IISc), Bengaluru

Aug 2022 – Jul 2027 (expected)

BS+MS (Research) in Physics

CGPA: 7.5/10

- Linear Algebra, Multi-variable Calculus, Probability & Statistics, Machine Learning, Matrix Theory, Stochastic models & Application, Mathematical Methods in Physics, Computational Physics.
- PhD-level coursework via Joint Astronomy Programme (IISc, IIA, RRI, ISRO): Radiative Processes, Galaxies & ISM, General Relativity & Cosmology, Astronomical Techniques, Fluid Mechanics & Plasma Physics

TECHNICAL SKILLS

- **ML / DL:** PyTorch, Deep Gaussian Processes, PINNs, Siamese networks, MC Dropout, heteroscedastic losses, MCMC, uncertainty quantification
- **Quantitative / Statistical:** Time-series inference on irregular & noisy data, bootstrap resampling, robust IQR statistics, Bayesian inference, signal reconstruction under stochastic noise, Probabilistic Modeling, Dynamic Programming.
- **Programming:** Python (NumPy, SciPy, Astropy, Matplotlib, Pandas, scikit-learn, etc), bash scripting; some C exposure; LaTeX.
- **HPC & Infrastructure:** Remeis Observatory cluster (SSH, SLURM-based workflows), parallel processing pipelines (23-worker), NuSTARDAS, XMM-SAS, XSPEC/ISIS

RESEARCH & ML EXPERIENCE

ML Researcher – GRB Light Curve Reconstruction

Apr 2025 – July 2026

National Astronomical Observatory of Japan (NAOJ) Advisor: Prof. Maria G. Dainotti

- Designed and implemented multiple deep learning architectures in PyTorch for irregularly sampled, noisy time-series reconstruction: Siamese network with learnable reference anchors, ReFANN (feedforward with heteroscedastic heads), Deep Gaussian Process.
- **Siamese network:** shared encoder, four-component fusion (query, reference, difference, Hadamard product $\rightarrow$ 128-dim), MC Dropout uncertainty ($p=0.1$, Gal & Ghahramani 2016), physics-informed composite loss (Gaussian NLL + $\lambda=10^{-3}$ monotonicity penalty).
- **ReFANN pipeline:** 522/545 GRBs processed (98.2%), 62% mean parameter uncertainty reduction vs. 39% benchmark, test MSE 0.0196; 5-fold CV, bootstrap resampling, z-score normalisation in log-space.
- **Deep Gaussian Process (DGP):** Implemented a two-layer stacked GP (GPyTorch, RBF kernel, Cholesky variational strategy, 35/25 inducing points) applied to 521/545 GRBs; achieved 11.6%, 12.3%, and 15.9% uncertainty reduction in $\log T_a$, $\log F_a$, and α respectively with α outlier rate of 1.92% — competitive with BNN while requiring no Monte Carlo weight sampling.
- Implemented MC Dropout convergence analysis ($n_{samples}=200$) and ablation studies across embedding dims (16/32/64) to empirically validate design choices.
- Applied robust statistics: IQR/1.35-based σ estimation, 3- σ MAD outlier rejection, spatial-correlation bootstrap ($\sigma=2.0$), 95% CI reporting across 544 GRBs.
- **Published:** Kaushal, Dainotti, Kumar et al. (2026), JHEAp. Code publicly available: github.com/Ritikkumar0730/GRB-Light-Curve-Reconstruction-2.
- Manuscript based on second paper is in preparation .

ML Researcher – Physics-Informed Neural Networks (PINNs)

Jan 2025 – Apr 2025

IISc Bengaluru Advisor: Prof. Rishita Das (Aerospace Engineering)

- Implemented PINNs in PyTorch to solve 1D/2D/3D Burgers' equations and 1D Euler equations (Sod shock tube) with sharp discontinuities; outperformed classical methods (Runge-Kutta, Euler, central difference) on discontinuity capture.
- Applied CFL criterion (CFL= 0.5, 400 grid points) for stable time-stepping; validated results against analytical solutions.

Lead Researcher – AGN Spectral Modelling (ECAP Collaboration)

Jan 2026 – Present

Erlangen Centre for Astroparticle Physics (ECAP), FAU Germany Full Group Member

- Full member of ECAP X-ray research group (granted Jan 2026 after independent international outreach); SSH access to Remeis Observatory HPC cluster.
- Lead researcher on testing new relativistic reflection models (relxill_ring, ring-corona geometry) against multi-instrument observational data; statistical model comparison via χ^2 fitting and MCMC. Manages multi-epoch, multi-instrument (XMM-Newton + NuSTAR) datasets; complete Python-based reduction pipelines on HPC.

Tracing of Outflows in Active Galactic Nuclei – Bachelor's Thesis

Sep 2025 – Apr 2026

Indian Institute of Astrophysics (IIA) Advisor: Prof. C. S. Stalin

- Built a Python pipeline to cross-match 1 million AGN catalogs (milliquas) with XMM-Newton archives (5 matching, SNR filtering); automated spectral extraction and multi-epoch analysis with parallel processing (23 workers) and finally ended to 26 sources.
- Discovered multi-phase ultra-fast outflows for the first time at 0.22c and 0.32c in a luminous quasar; independently developed custom XSTAR photoionisation grids – no public model could reproduce the soft X-ray signatures.
- **Manuscript in preparation** targeting *MNRAS*; collaborating with Prof. James Reeves (UMBC).

Spectral Analysis of Seyfert Galaxies

Dec 2024 – Jan 2026

Indian Institute of Astrophysics (IIA) Advisor: Prof. C. S. Stalin

- Developed Python cross-match 30,000 entries, reduction pipeline including - calibration, background filtering, and signal to noise criteria.
- Pipeline for NuSTAR/XMM-Newton multi-epoch data; broadband spectral modelling to measure coronal temperatures (using XSPEC) for multiple AGN. Reported coronal temperature measurements for multiple sources for the first time.
- **Manuscript in preparation.**

PUBLICATIONS

Published

- Kaushal, A., Dainotti, M.G., **Kumar, R.**, et al. (2026). Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves. *Journal of High Energy Astrophysics (JHEAp)*. DOI: [10.1016/j.jheap.2025.100519](https://doi.org/10.1016/j.jheap.2025.100519)

In Preparation

- **Kumar, R.**, et al. Light Curve Reconstruction of GRBs using Machine Learning. Second paper in NAOJ series (lead ML contributor).
- **Kumar, R.**, Stalin, C.S., Reeves, J.N. Multi-phase Ultra-fast Outflows in a Luminous Quasar. Target: *MNRAS*.
- **Kumar, R.**, Stalin, C.S., Hota, J. Variation in Coronal Temperature in Seyfert-type AGN. In preparation, 2026.

CONFERENCES & WORKSHOPS

- XRISM International Conference 2025, Japan – [Certificate](#).
- Astronomical Society of India Conference 2026, Guwahati – ([Certificate](#), [Poster](#)).
- Fast Radio Bursts Workshop, ICTS India.

ACHIEVEMENTS

- **Research Internship, NAOJ Tokyo (Summer 2025)** – with Prof. Maria G. Dainotti
- **Full membership, ECAP X-ray Research Group, Germany** since Jan. 2026
- **Research collaboration with the Indian Institute of Astrophysics (IIA), Bengaluru** since Dec. 2024.
- **Management Coordinator**, Pravega 2024–25 (India's largest science & cultural festival), IISc